

# Final Individual Project

## One-page Summary

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### Research question

Is there a significant difference in mean height among students with different levels of milk drinking frequency?

### Explanatory variable

- Explanatory variable: NoMilkDrinking
- Definition: This variable classifies students into three groups for analysis, where 0 = No milk drinking, 1 = Occasional milk drinking, and 2 = Frequent milk drinking.

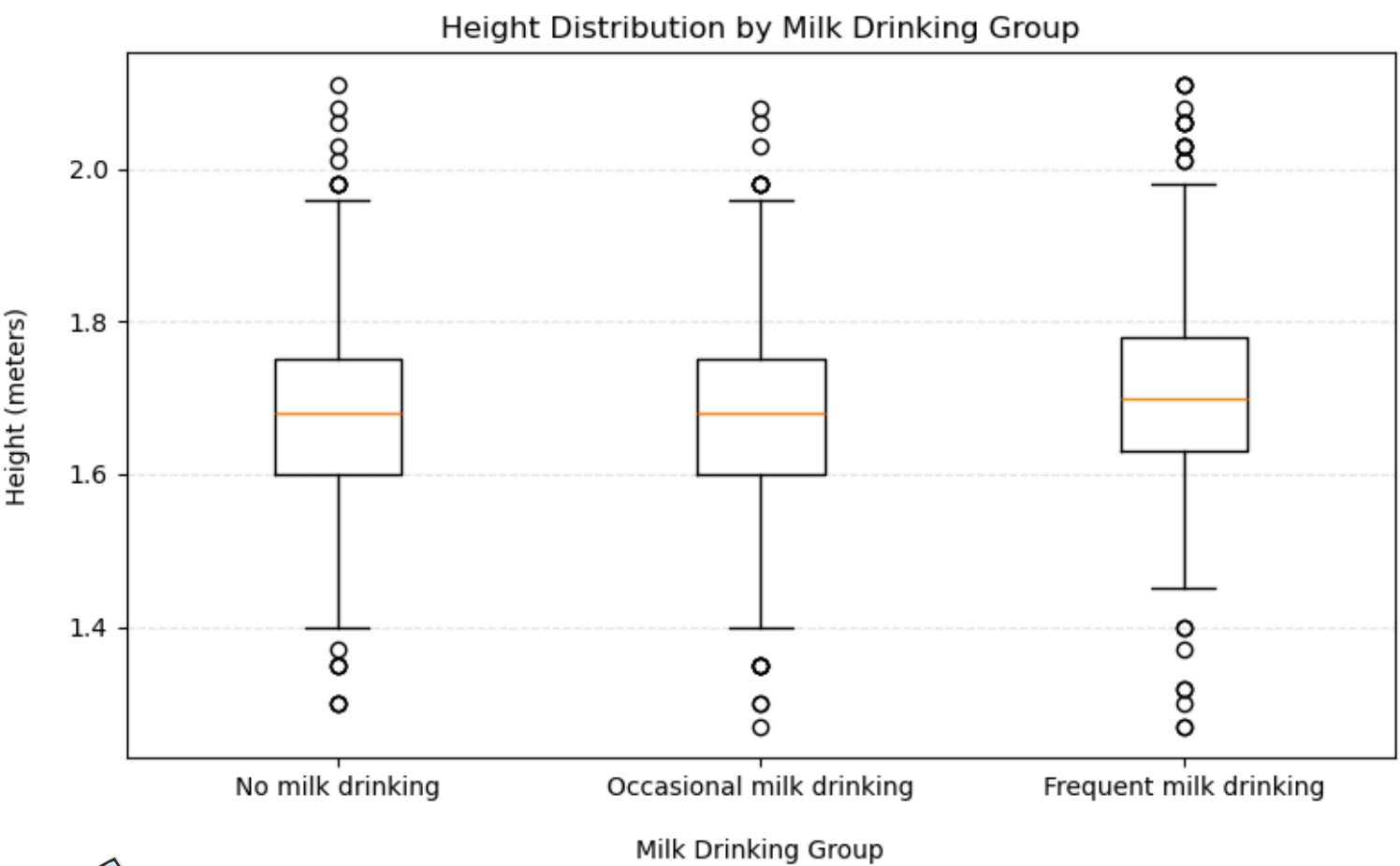
### Response variable

- Response variable: HowTallAreYouWithoutShoesInMeters
- Definition: This variable records students' height without shoes in meters.

### Method

- 1.One-Way ANOVA
- 2.Tukey HSD post-hoc test

### Figure



### Table

| Comparison   | Group 1                      | Group 2                      | Mean Difference |
|--------------|------------------------------|------------------------------|-----------------|
| Comparison 1 | 0 = No milk drinking         | 1 = Occasional milk drinking | 0.0104          |
| Comparison 2 | 0 = No milk drinking         | 2 = Frequent milk drinking   | 0.0364          |
| Comparison 3 | 1 = Occasional milk drinking | 2 = Frequent milk drinking   | 0.0259          |

### Key result

- The null hypothesis ( $H_0: \mu_1 = \mu_2 = \mu_3$ ) was rejected at  $\alpha = 0.05$ .
- The one-way ANOVA indicated that at least one group mean height differed among the three groups ( $F(2, 12665) = 139.056, p < 0.001$ ).
- Tukey HSD post hoc tests showed significant pairwise differences among all groups.
- Descriptive statistics indicated that frequent milk drinkers had the highest mean height, followed by occasional milk drinkers and non-drinkers.

### Conclusion

There is sufficient statistical evidence at the 0.05 significance level to conclude that at least one group mean height differs among the three milk drinking groups.